PHYSICS

Time: 3 Hours **Total Marks: 70**

Note:

i. All questions are compulsory.
ii. Neat diagrams must be drawn wherever necessary.
iii. Figures to the right indicate full marks.
iv. Use of only logarithmic table is allowed.
v. All symbols have their usual meaning unless otherwise stated.
vi. Answers to both sections must be written in the same answerbook.
vii. Answer to every question must be written on a new page.

SECTION – I

Q.1. Select and write the most appropriate answer from the given alternatives for each sub-question: **[7]**

i. In simple harmonic motion, the acceleration of a particle is zero when its _____.
(A) velocity is zero
(B) displacement is zero
(C) both velocity and displacement are zero
(D) both velocity and displacement are maximum

ii. As we move from the equator towards the pole, weight of a body _____.
(A) remains the same (B) becomes zero
(C) decreases (D) increases

iii. Substances which break just after their elastic limit is reached are _____.
(A) ductile (B) brittle
(C) malleable (D) plastic

iv. The dimensions of angular momentum are _____.
(A) $[L^{-2} M^1 T^{-1}]$ (B) $[L^2 M^1 T^{-1}]$
(C) $[L^1 M^2 T^1]$ (D) $[L^2 M^2 T^{-2}]$

v. The number of degree of freedom for a rigid diatomic molecule is _____.
(A) 3 (B) 5
(C) 6 (D) 7

vi. A wave is travelling in the negative direction of X-axis, then its equation is _____.
(A) $y = a \sin 2\pi\left(\frac{t}{T}-\frac{x}{\lambda}\right)$ (B) $y = a \sin 2\pi n \left(t+\frac{x}{v}\right)$
(C) $y = a \sin 2\pi\left(\frac{t}{T}+\frac{x}{v}\right)$ (D) $y = a \sin \frac{2\pi}{T}\left(t+\frac{x}{\lambda}\right)$

vii. A rectangular film of a liquid is 5 cm long and 4 cm wide. If the work done in increasing its area to 7 cm × 5 cm is 0.06 J, the surface tension of the solution is:
(A) 0.02 J/ m^2 (B) 0.2 J/ m^2
(C) 2 J/ m^2 (D) 20 J/ m^2

Q.2. Attempt any SIX: **[12]**

i. What is geo-stationary satellite? State its any 'two' uses.
ii. Define radius of gyration and give its physical significance.
iii. What is capillarity? Give any 'two' applications of capillarity.
iv. Draw a neat and labelled diagram of Ferry's black body.

v. The frequency of revolution of a particle performing circular motion changes from 60 r.p.m. to 180 r.p.m. in 20 seconds. Calculate the angular acceleration of the particle. ($\pi = 3.142$)

vi. Find the altitude at which the acceleration due to gravity is 25% of that at the surface of the earth. (Radius of the earth = 6400 km)

vii. A simple pendulum of length 1 m and mass 10 g oscillates freely with amplitude 2 cm. Find its potential energy (P.E) at the extreme position. ($g = 9.8\ m/s^2$)

viii. The length of an air column for a fundamental mode in a resonance tube is 16 cm and that for second resonance is 50.25 cm. Find the end correction.

Q.3. Attempt and THREE: **[9]**

i. Show that beats frequency is equal to frequency difference between two interfering waves.

ii. Derive an expression for strain energy. Hence show that strain energy per unit volume is directly proportional to the square of the stress.

iii. Compare the rates of loss of heat by a black body at 627 °C and 327 °C, if temperature of surrounding is 27 °C.

iv. A metre gauge train is moving at 72 km/ hr along a curved rail-way of radius of curvature 500 m at a certain place. Find the elevation of the outer rail above the inner rail so that there is no side pressure on the rail. ($g = 9.8\ m/s^2$)

Q.4. A. Define an ideal simple pendulum. Show that motion of a simple pendulum under certain conditions is simple harmonic. Obtain an expression for its period.

B. A solid sphere of diameter 50 cm and mass 25 kg rotates about an axis through its centre. Calculate its moment of inertia. If its angular velocity changes from 2 rad/s to 12 rad/s in 5 seconds, calculate the torque applied. **[7]**

OR

A. Draw three neat diagram for modes of vibration of an air column in a pipe, when

i. the pipe is open at both ends, and

ii. the pipe is closed at one end.

Hence derive an expression for fundamental frequency in each case.

B. A horizontal circular loop of a wire of radius 0.02 m is lowered into crude oil and a film is formed. The force due to the surface tension of the liquid is 0.0113 N. Calculate the surface tension of the crude oil. ($\pi = 3.142$) **[7]**

SECTION – II

Q.5 Select and write the most appropriate answer from the given alternatives for each sub-question: **[7]**

i. When a ray of light enters into water from air, _____.

(A) its wavelength decreases (B) its wavelength increases

(C) its frequency increases (D) its frequency decreases

ii. The condition for destructive interference between two waves is that their phase difference should be _____.

(A) $0, \pi, 2\pi, \ldots.$ (B) $0, 2\pi, 4\pi, \ldots.$

(C) $0, 3\pi, 5\pi, \ldots..$ (D) $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots.$

iii. An ideal ammeter has _____.

(A) low resistance (B) high resistance

(C) infinite resistance (D) zero resistance

iv. Which logic gate corresponds to the logical equation $Y = \overline{A+B}$?

(A) NAND (B) NOR

(C) AND (D) OR

v. In a photocell, increasing the intensity of incident light increases _____.
(A) the stopping potential
(B) the photoelectric current
(C) the energy of the incident photons
(D) maximum kinetic energy of the photo-electron

vi. In series LCR circuit R = 3 Ω, X_L = 8 Ω, X_C = 4 Ω, the impedance of the circuit is:
(A) 3 Ω (B) 7 Ω
(C) 5 Ω (D) 25 Ω

vii. A microphone which converts sound into electrical signal is an example of _____.
(A) a thermister (B) a rectifier
(C) a modulator (D) an electrical transducer

Q.6 Attempt any six: **[12]**

i. State Gauss' theorem in electrostatics. State the expression for electric field intensity at a point outside an infinitely long charged conducing cylinder.
ii. What is Curie temperature? What happens above Curie temperature?
iii. Draw a neat and labelled diagram of earth's atmosphere.
iv. The optical path difference between two identical waves arriving at a point 371λ. Is the point bright or dark? If the path difference is 0.24 mm, calculate the wavelength of light used.
v. State Kirchhoff's laws in electricity.
vi. In a potentiometer the balancing length of the wire is found to be 2.5 m for a cell of e.m.f. 1.5 V. Find the balancing length of the wire for another cell of e.m.f. 1.2 V on the same potentiometer.
vii. A closely wound solenoid of 1000 turns and area of cross-section 2×10^{-4} m^2 carries a current of 1 A. It is placed in a horizontal plane with its axis making an angle of 30° with the direction of uniform magnetic field of 0.16 T. Calculate the torque acting on the solenoid.
viii. The energy of an excited hydrogen atom is –0.85 eV. Find the angular momentum of the electron. ($h = 6.63 \times 10^{-34}$ J.s., $\pi = 3.142$, $E_1 = -13.6$ eV)

Q.7 Attempt any three: **[9]**

i. Explain the working of a transistor as a switch.
ii. State Einstein's photoelectric equation. Explain any two characteristics of photoelectric effect on the basis of this equation.
iii. Determine the change in wavelength of light during its passage from air to glass, if the refractive index of glass with respect to air is 1.5 and frequency of light 4×10^{14} Hz. ($c = 3 \times 10^8$ m/s)
iv. In Young's double slit experiment, the slits are 0.5 mm apart and interference is observed on a screen placed at a distance of 100 cm from the slit. It is found that 9th bright fringe is at a distance of 8.835 mm from the 2nd dark fringe, on the same side of the centre of the fringe pattern. Find the wavelength of light used.

Q.8 A. What is series LCR resonant circuit? Obtain the expression for impedance. Hence state the condition for series resonance and derive the expression for resonant frequency.
B. A 10 μF capacitor is connected with 100 V battery. What would be the electrostatic energy stored? **[7]**

OR

Q.8. A. Obtain an expression for energy of an electron in Bohr orbit. Hence obtain the expression for its binding energy.
B. A rectangular coil having 100 turns each of length 1 cm and breadth 0.5 cm is suspended in radial magnetic induction 0.002 T. The torsional constant of suspension fiber 2×10^{-8} Nm/degree. Calculate the current sensitivity of a moving coil galvanometer. **[7]**